

Exercise project 1

$$d = 3$$

$$1) f = 3x_1 + \frac{100}{x_1 x_2} + 3x_2$$

(1)

$$\frac{\partial f}{\partial x_1} = 3 - \frac{100}{x_1^2 x_2} \quad \frac{\partial f}{\partial x_2} = 3 - \frac{100}{x_1 x_2^2}$$

$$(2) \nabla f = 0$$

$$3 = \frac{100}{x_1^2 x_2} \Rightarrow x_1^2 x_2 = \frac{100}{3}$$

$$3 = \frac{100}{x_1 x_2^2} \Rightarrow x_1 x_2^2 = \frac{100}{3}$$

(3) Solve system

$$x_1^2 x_2 = x_1 x_2^2 \Rightarrow x_1 x_2 (x_1 - x_2) = 0$$

We have to consider that x_1 and x_2 ~~don't~~ can't be zero by $\frac{100}{x_1 x_2}$
the unique solution is that x_1 and x_2 are equal to set zero in $x_1 - x_2$.

$$\triangleright x_1 = x_2$$

substitute

$$x_1^3 = \frac{100}{3} \Rightarrow x_1 = \left(\frac{100}{3}\right)^{1/3} \approx 3.2075$$

$$\text{Stationary point} = x^{(1)} = (3.2075, 3.2075)$$

(4) Hessian

$$\frac{\partial^2 f}{\partial x_1^2} = \frac{200}{x_1^3 x_2}, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{100}{x_1^2 x_2^2}, \quad \frac{\partial^2 f}{\partial x_2^2} = \frac{200}{x_1 x_2^3}$$

~~or~~ We know that x_1 and x_2 are equal, so each result have x^4 on the denominator.
First main minor: $200 > 0$
Determinant: $200^2 - 100^2 > 0$

$$H(x_1, x_2) = \frac{1}{x^4} \begin{pmatrix} 200 & 100 \\ 100 & 200 \end{pmatrix} \text{ therefore } x^{(1)} \text{ is a local minimum}$$

$\hookrightarrow$ Where $x^4 \rightarrow$ include x_1 and x_2